TopicForest: Embedding-driven Hierarchical Clustering and Labeling for Biomedical Literature

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Introduction. Navigating the growing biomedical literature requires topic modeling methods that go beyond flat clustering. Existing models like Top2Vec[1] and BERTopic[2], while effective, do not capture the hierarchical and nested structure of scientific knowledge. Meanwhile, traditional hierarchical models rely on outdated bag-of-words approaches, ignoring advances in embedding-based semantic representation. We propose TopicForest, a novel framework that constructs, labels, and visualizes multi-scale topic hierarchies over large biomedical corpora using LLM-derived embeddings and clustering.

Methods & Materials. TopicForest integrates four core components: (1) Document Embedding: Titles and abstracts are embedded using gte-large-en-v1.5 to produce 1024-dimensional vectors. (2) Dimension Reduction: Embeddings are mapped to 2D using UMAP or t-SNE for interpretability. (3) Hierarchical Clustering: A dendrogram is built via Ward's method and "cut" at multiple levels to yield fine-to-coarse topic layers, as illustrated in Figure 1(a). (4) LLM-Based Labeling: To generate interpretable topic labels across the hierarchy, TopicForest applies an LLM-driven strategy that adapts to the cluster's structural role. As illustrated in Figure 1(b), Leaf clusters are labeled by summarizing document titles into concise names and descriptions. Deriving clusters inherit labels from their single child to maintain semantic continuity. Merging clusters, which have multiple children, are labeled by prompting the LLM to synthesize a unifying theme from their child labels. This structure-aware labeling approach ensures consistency across levels and yields higher diversity and semantic coherence than keyword-based methods like c-TF-IDF[2]. Using 24,336 biomedical articles from Scientific Reports, we benchmark clustering against human-labeled category trees. We measure Adjusted Mutual Information (AMI) for clustering quality and Diversity and Hierarchical Affinity (HA) for labeling quality. Our baseline is BERTopic with c-TF-IDF for labeling.

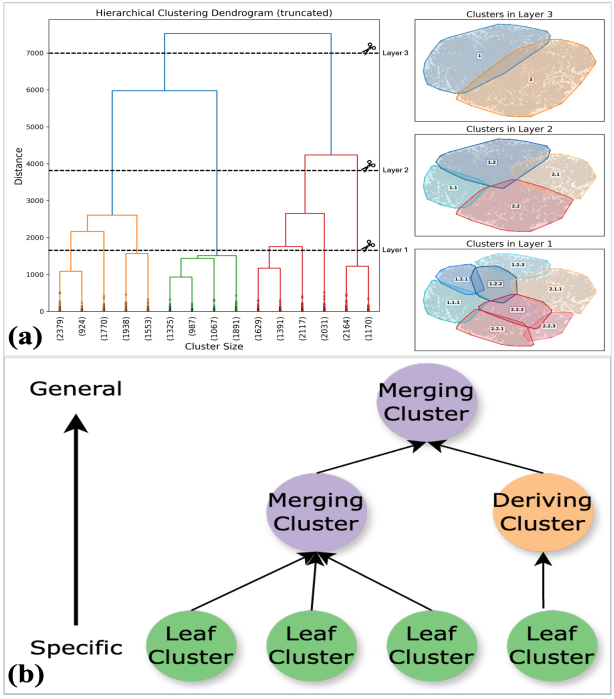

Figure 1, (a) Illustration of dendrogram cutting for multi-layered topic hierarchy and (b) Three different types of clusters in the topic hierarchy.

Key Findings. TopicForest matches or exceeds BERTopic in clustering quality across all layers, offering multi-resolution clustering via a single dendrogram. LLM-based labels show higher diversity and semantic coherence than c-TF-IDF (e.g., HA increases from ~0.3 to ~0.6 – 0.67). Visualization confirms TopicForest's ability to separate broad themes and granular subtopics.

Conclusions & Future work. TopicForest unifies semantic embeddings and LLM-based labeling in a scalable, interpretable framework for hierarchical topic modeling. It enables dynamic, multi-resolution literature exploration and reveals latent biomedical themes. Future work will enhance dimensionality from 2D to 3D, optimize hierarchy construction, and scale to full PubMed corpora.

References

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